Reading Greek Squeezes with Page Networks

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Abstract

A paper squeeze records an inscription as an impression of the carved stone. Its letters appear in light and shadow, alongside folds, cracks and ornament. We describe a system that reads two photographs of a squeeze, lit from different directions. It aligns the photographs, combines the visible strokes, and reads each possible letter together with the marks around it. Models of Greek letter sequences help choose between uncertain readings; they cannot fill a gap where no letter was detected. Five versions of the processing chain produce transcripts that are aligned and voted on. We follow the images through these steps and show both successful readings and remaining errors. The character error rate on confidently annotated text is 9.59% on 224 public squeezes, with each squeeze held out from its reader's training and from the choice of its weights, and 9.23% for the fixed final output on 50 TROGS-26 test squeezes.

1. What has to be read

Wet paper pressed into an inscribed stone and dried keeps an impression of the carving. Photographing the squeeze with light grazing its surface makes the relief visible. The TROGS collection [1] provides two such photographs, R1 and R2, with lighting directions roughly a right angle apart. A stroke across the light shows strongly; one along it may almost disappear. A shadow is therefore evidence about surface shape, not a dark pixel that can be read as ink. The same light also reveals paper grain, tears, folds, breaks in the stone and carved ornament.

Figure 1 shows the problem on public squeeze IG_II2_1973a, the *running squeeze* of later figures. Its rows share the page with wreaths and a paper label. The *reader*, the part of the system that decides what each mark is (Section 6), must find a faint letter without copying the label or reading the wreath as text. It must distinguish two questions: *is this region a letter?* and *which letter is it?* An isolated crop often cannot answer either. The size of the mark, its alignment with a row, a second box on the same stroke, and the letters around it all matter.

The output has one line per inscription row, without word spaces, translation or restored text. The dataset writes each Greek letter as one Latin letter:

Α	Β	Γ	Δ	Ε	Ζ	Η	Θ	Ι	Κ	Λ	Μ	Ν	Ξ	Ο	Π	Ρ	Σ	Τ	Υ	Φ	Χ	Ψ	Ω
A	B	G	D	E	Z	H	Q	I	K	L	M	N	X	O	P	R	S	T	Y	F	C	U	W

Three signs, `&`, `:` and `)`, complete its 27 symbols; `:` marks an interpunct. Transcripts in this report use these symbols, as the dataset and its scorer do. Where letter shapes matter, as in the confusions of Section 8, letters are written in Greek. Every output character must come from a detected region and be one of the reader's proposed letters for that region.

We measure errors against the supplied reference transcripts. The main measure, *Core character error rate* (Core CER), counts missing, wrong and extra characters against confidently annotated text, including row boundaries. A lower rate is better. Section 12 explains the scoring and training exclusions in full.

2. From two photographs to one transcript

The two photographs cannot simply be overlaid. The paper may have moved or deformed, and stitched scans may disagree locally. The first task is to align the same piece of paper in both views. Their shadows then provide two estimates of surface slope, which we render under fixed lights so that the reader sees a consistent presentation of the strokes (Figure 2).

A *detector* draws a rotated box around each possible letter. We keep its uncertain proposals: discarding a faint letter here would make it unavailable to the rest of the system. Each box becomes a *token*: an image crop together with its position, shape and detection scores.

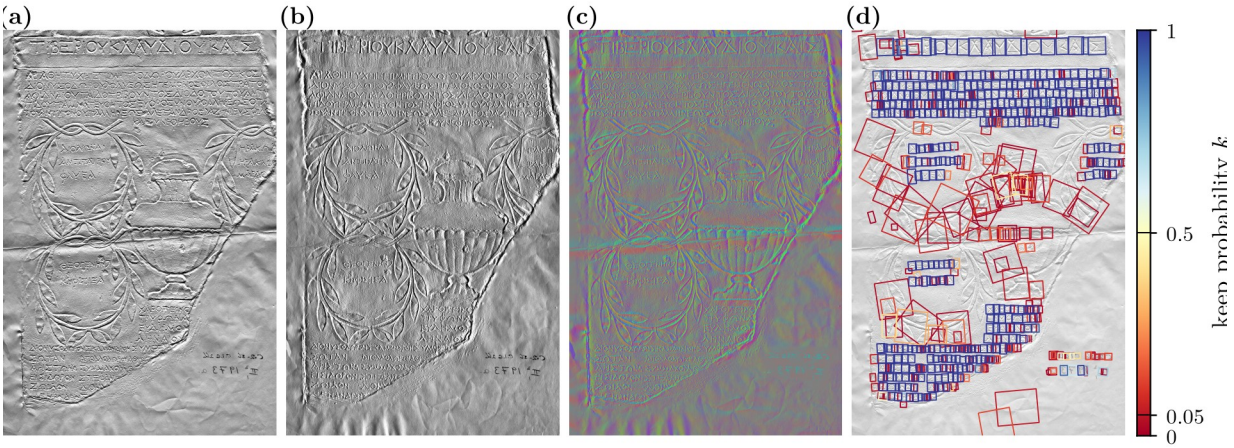


Figure 1: Letters, folds and ornament under changing light. Public squeeze IG_II2_1973a. (a, b) The two photographs. (c) The combined relief under three fixed lights, one per colour channel. (d) All 707 candidate boxes, coloured by the reader's probability of keeping them as letters: blue is high, red low. The ornament attracts many boxes but few accepted letters. This squeeze was excluded from its reader's training; its 4.48% Core CER is better than the public average.

Two *page networks* read these tokens; together they form the reader, whose probabilities combine both (Section 7). Unlike a classifier that sees only one crop, each network also sees nearby marks and their arrangement on the page. It assigns probabilities to the 27 symbols and to *none*, meaning that the box should not contribute a letter. The probability of any letter rather than *none* is its *keep probability*, denoted k.

The *decoder* turns those probabilities into text. It groups likely letters into rows, lets weaker marks join nearby rows, and compares possible readings. A reading can use one of a token's proposed letters or skip that token. Character models, learned from public transcripts and open Greek text, help rank these alternatives. Repeated names and phrases in the squeeze's other rows provide further context.

On a new squeeze, five separately trained detectors produce five versions of this process, called *chains*. They share the same page networks and text models, but may find different boxes or align the images differently. Their complete transcripts are aligned and combined by a vote (Figure 3). For public evaluation, we instead use one chain whose models were trained without the squeeze being read. The public set is divided into five groups, or *folds*, for this purpose.

3. Aligning the two photographs

3.1 Why brightness cannot simply be matched

For a locally matte surface with height $h(x, y)$, lit from direction ϕ at low elevation θ, the brightness relative to the surrounding paper is approximately

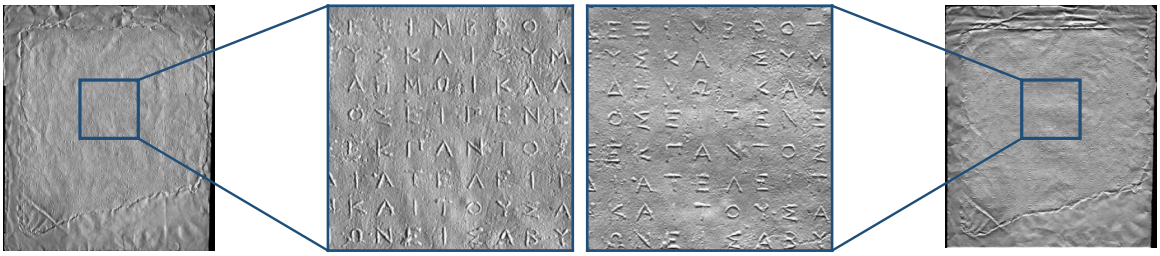
$$s = \frac{I}{I_{\text{paper}}} - 1 \approx -\cot\theta\,(\cos\phi\, h_x + \sin\phi\, h_y). \tag{1}$$

Here ϕ is the light's angle in the plane of the paper and θ its angle above that plane. The derivatives h_x and h_y describe surface slope in the two image directions. I_{paper} is a smooth estimate of the local paper brightness; dividing by it suppresses broad illumination changes. A photograph measures a directional derivative of height. The two raw images are therefore not two brightness-preserving views of the same pattern: changing the light changes which strokes are bright or dark.

In this collection, one view's light runs approximately along x and the other's along y. Which is which is estimated from directional gradient energy. Call their relative shadings s_x and s_y.

1 Photograph under two lights

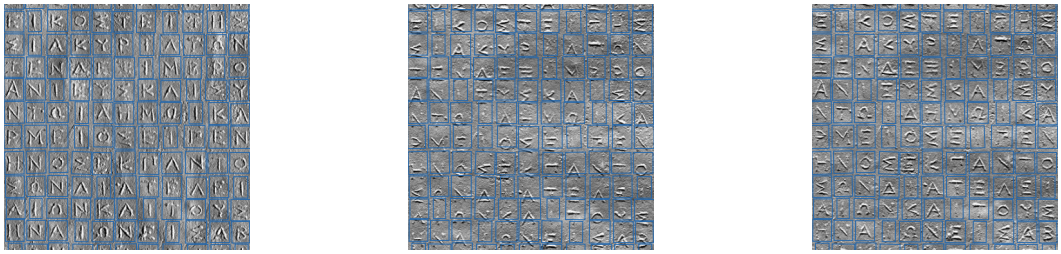
The same squeeze, with different strokes revealed by each light.



R_1: lit along y | R_1, enlarged | R_2, enlarged | R_2: lit along x

2 Align the paper

Match the same locations in both photographs, including local distortions.



R_1: fixed coordinates | R_2: global alignment | R_2: local alignment

3 Locate possible letters

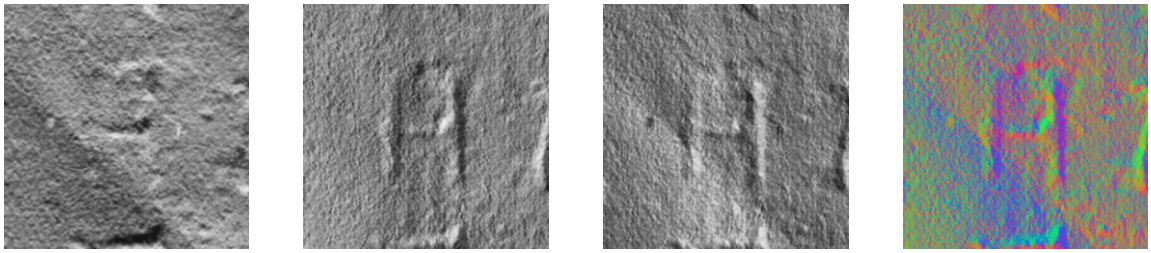
Keep uncertain boxes so that the reader can judge them in context.



656 candidate boxes | Detail: a crop and position for each box

4 Show the strokes under fixed lights

Combine the two views into a consistent colour image for the reader.

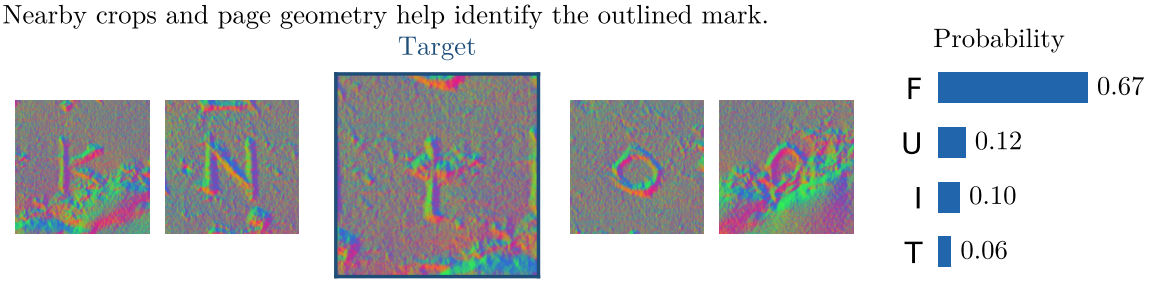


light from 90° | light from 210° | light from 330° | Combined crop

Figure 2: From photographs to letter crops. All images show public squeeze IG_II2_493_Copy1. The enlargements reveal the different strokes visible under each light. Alignment brings them into correspondence; detection locates possible letters; relighting combines their evidence into the colour crop read by the network. Figure 3 continues from here.

5 Read in context

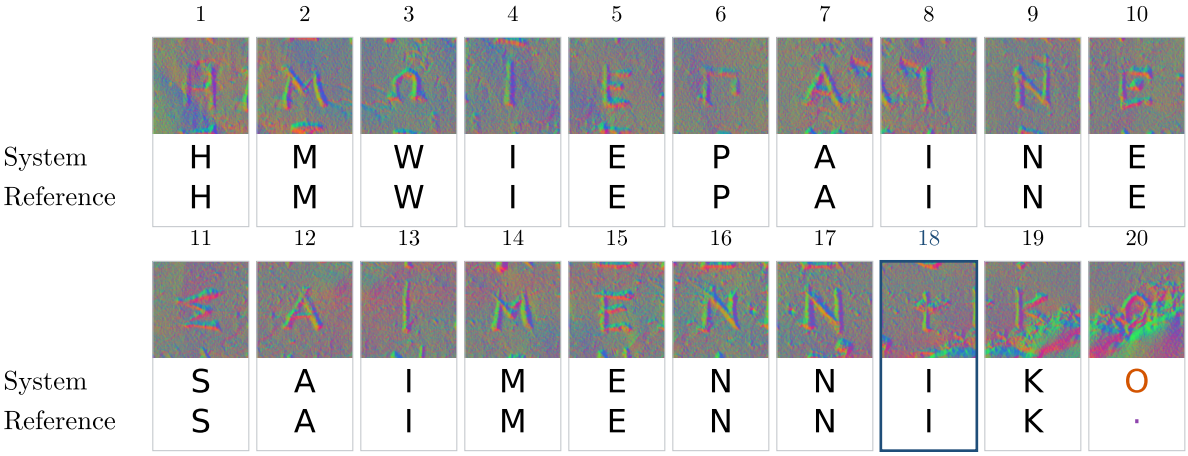
Nearby crops and page geometry help identify the outlined mark.



Four of the target's 32 neighbours are shown.

6 Choose a reading

Public row 23: compare the system's letters with the reference.



Blue outline: F corrected to I. Orange: an extra O remains.

7 Align the readings, then vote

Test output: IAS_27146, row 9. Chain 3 is the pivot.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26
Chain 1	C	A	S	A	N	T	O	·	L	E	·	·	E	N	A	S	M	H	D	E	K	E	K	A	L	Y
Chain 2	C	A	S	A	M	·	·	·	L	E	·	·	E	N	A	S	M	H	D	E	K	E	K	A	L	Y
Chain 3	C	A	S	A	M	P	E	P	L	E	G	M	E	N	A	S	M	H	D	E	K	E	K	A	L	Y
Chain 4	C	A	S	A	M	P	E	P	L	E	G	M	E	N	A	S	M	H	D	E	K	E	K	A	L	Y
Chain 5	C	A	S	A	M	·	·	·	L	E	·	·	E	N	A	S	M	H	D	E	K	E	K	A	L	·
Vote	C	A	S	A	M	P	E	·	L	E	·	·	E	N	A	S	M	H	D	E	K	E	K	A	L	Y

Position 6: gap 2, P 2, T 1. The tie keeps pivot P.

Position 8: gap 3, P 2. The gap wins; no letter is emitted.

Figure 3: From letter crops to a transcript. Stages 5–6 use public squeeze IG_II2_493_Copy1; its reference is shown beneath the system's letters. Orange marks an extra output letter. Stage 7 shows the actual outputs of the five test chains. Each cell is one aligned position; red marks disagreement with the vote and purple dots are gaps, which emit no letter. The pivot is the transcript closest to the other four, measured by the total number of edits needed to align them. Section 11 gives the voting rules.

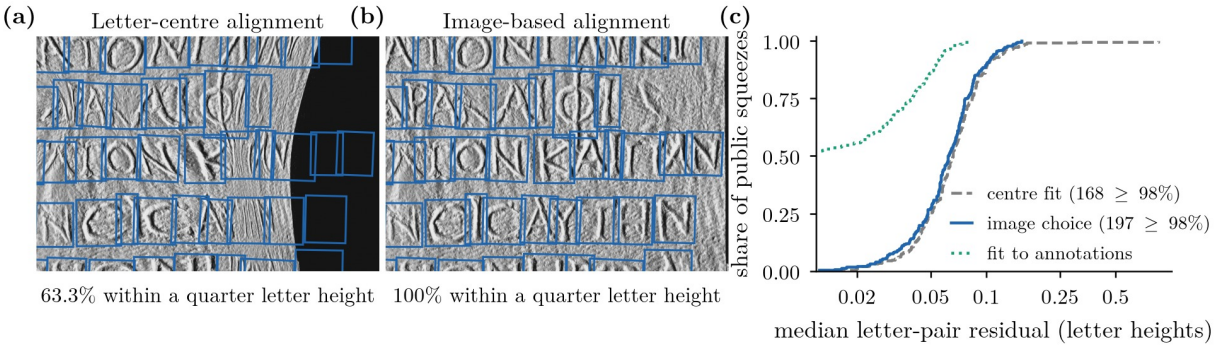


Figure 4: Correcting a local alignment error. Public squeeze IAS33163, in the region where the two maps disagree most. (a) Alignment from detected letter centres. (b) Alignment selected from image agreement. The boxes stay fixed in R1 coordinates. Over the whole squeeze, the share of annotated letter pairs within a quarter letter height rises from 63.3% to 100%. (c) Across the public set, cumulative distributions of median alignment error. The annotation-fitted curve is a reference comparison only; those correspondences are not supplied to the system. Legend counts use the stricter 98%-within-a-quarter-height criterion of Appendix A.1.

Apart from scale and sign, they estimate h_x and h_y. Differentiate each *across* its light:

$$\partial_y s_x \;\propto\; h_{xy} \;\propto\; \partial_x s_y. \tag{2}$$

The two derivative images now describe the same change in surface slope. Their magnitudes can therefore guide alignment even where different letters are visible in the two views. This is an approximation: shadows, non-matte paper and errors in lighting direction need not obey Eq. (1). The method checks local image agreement rather than assuming a perfect physical model.

3.2 Matching the whole page, then local regions

A coarse detector pass on each view estimates letter scale and offers a map from detected centres. That map can include a global transformation, separate fits across stitched scans and a smooth local deformation (a thin-plate spline [5]). But letters visible in one photograph may be missing in the other, so centre matching alone can accept the wrong correspondence.

A second route registers the mixed-derivative images. Phase correlation searches quarter turns and small angular offsets; enhanced correlation coefficient alignment [6] fits Euclidean and affine maps. Three local levels then refine the map tile by tile, each allowing smaller displacements (Table 3). Accepted shifts are fitted with a thin-plate spline, smoothed until its orientation is preserved. The centre-derived map is also refined with the two finer local levels. Median tile correlation then selects the best of these three maps. No annotation is used in that selection. We express all box positions and crops in R1 coordinates and warp R2 to match. This choice makes the implementation asymmetric: the two views are not interchangeable.

Figure 4 separates the image-based decision from its evaluation. The map is chosen on pixels; annotated letter correspondences, available on the public set, independently measure the residual. Appendix A.1 counts the squeezes this improves. A small median residual does not guarantee that every stitched seam is correct.

4. Combining the light from both views

After registration, each shading is normalised by its root mean square on the paper. Mixed derivatives set the relative sign of the two shadings and support a final tilewise correction of residual alignment. Their local correlation ρ controls how much of view 2 to trust:

$$\alpha = \mathrm{clip}\left(\frac{\rho - 0.1}{0.2}, 0, 1\right), \qquad (g_x, g_y) = (\hat{s}_x, \sigma\alpha\hat{s}_y) \tag{3}$$

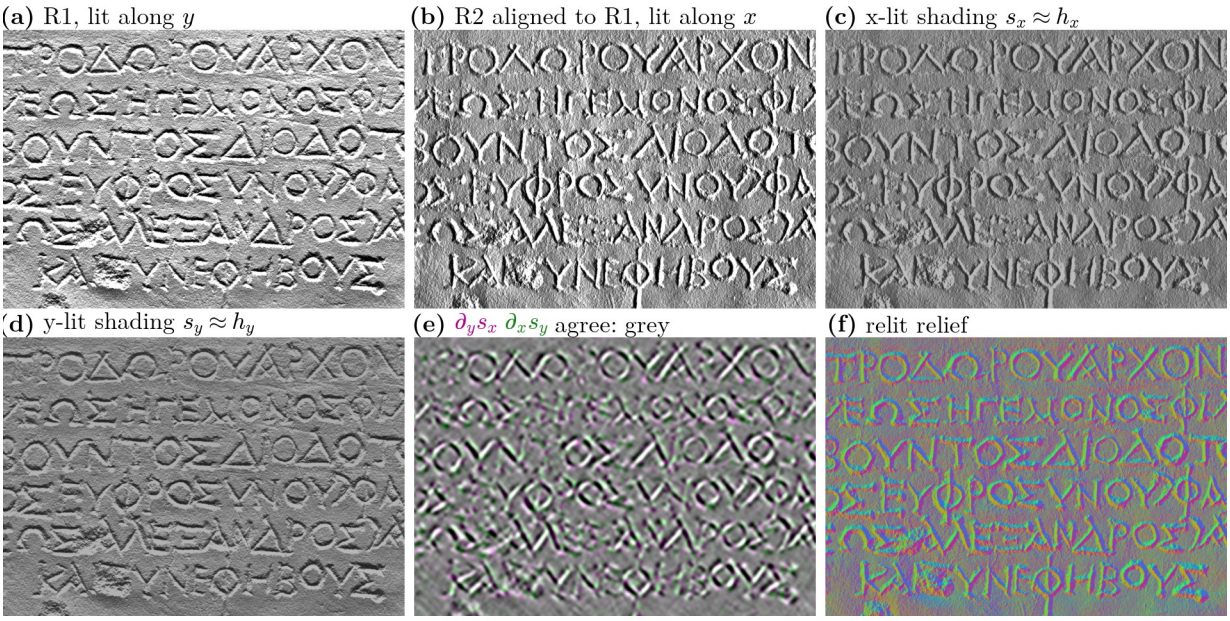


Figure 5: From shadows to a common lighting representation. A 10.5 by 7.3 letter-height window of IG_II2_1973a. (a) R1, lit along y. (b) R2 registered onto it, lit along x. (c, d) Their relative shadings: up to scale and sign, h_x and h_y. Grey denotes zero. (e) The two mixed derivatives: one in magenta, the other in green; agreement is grey, disagreement tinted. Their correlation is 0.92 in this window and 0.72 over the page. (f) The combined field under the three fixed lights. All panels depict the same paper; the reader's crops are cut from the final panel.

when R1 is the x-lit view; the attenuated component is exchanged when it is y-lit. σ is the relative sign. Where the scans show different paper or cannot be aligned, the representation falls back to the component supplied by R1. This avoids combining unrelated strokes, at the cost of losing the second light there. The absolute groove/ridge ambiguity is resolved by choosing the sign whose curvature inside confident detections has the profile of carved strokes.

The system uses these slopes directly, without reconstructing the surface's absolute depth. Three fixed virtual lights, at azimuths $\phi = 90°$, 210° and 330°, produce one image channel each:

$$r_\phi = -(\cos\phi\, g_x + \sin\phi\, g_y), \qquad R_\phi = 128 + 127\,\mathrm{clip}\left(\frac{r_\phi}{4 \cdot 1.4826\ \mathrm{med}\,|r_\phi|}, -1, 1\right). \tag{4}$$

Each colour channel is one light direction; colour carries no other meaning. A stroke invisible to one virtual light can show in another. The rendering standardises the evidence; it does not create relief absent from both photographs.

5. Finding possible letters

The detector is YOLO26-L with oriented boxes [4], initialised from DOTA aerial-image weights and fine-tuned for one class: a possible letter. Each of five detectors trains on the portion of the 181-squeeze training set outside its fold. Training includes each view separately and a combined image whose channels are R1, registered R2 and their pixelwise minimum. Rotation reaches 60° and scale ranges from 0.6 to 1.6. A common checkpoint rule selects the weights after training pass (epoch) 59; details are in Appendix A.

When reading a page, a coarse pass first measures letter size and refines alignment. The detailed pass rescales the page to 40-pixel letters and cuts it into 1280×1280 tiles with 25% overlap. We measure the overlap of two boxes by intersection over union (IoU): the area they share divided by the area covered by either. Within each tile of one image, rotated non-maximum suppression (NMS) keeps the better-scoring of two boxes that overlap at IoU above 0.8. A box belongs to its

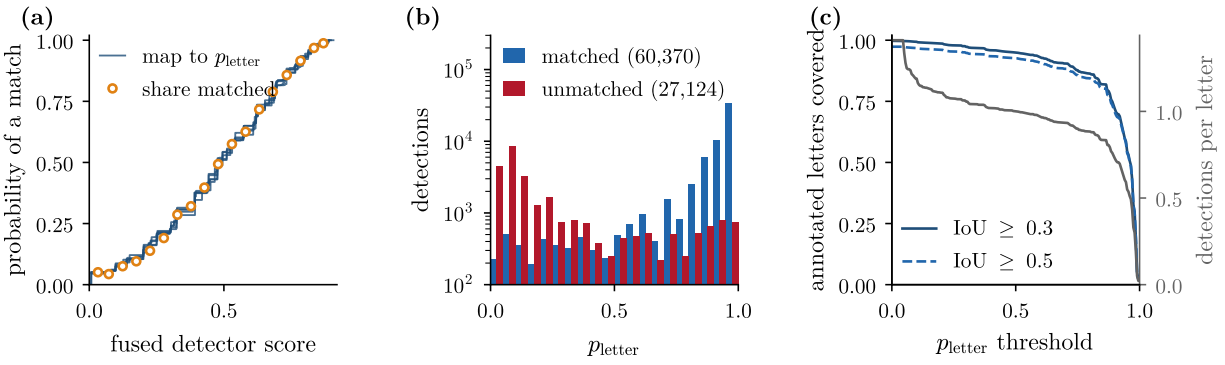


Figure 6: A low detector score need not mean a false letter. Held-out detections on 224 public squeezes. (a) Calibration curves and observed match frequencies. (b) Scores of matched and unmatched boxes; unmatched boxes include duplicates. (c) The fraction of annotated letters reached by at least one box above a score threshold, at two overlap requirements. Grey shows the number of boxes remaining per annotated letter. The reader receives all 87,494 proposals.

nearest tile centre, avoiding repeated tile-edge outputs. Detections from the three images are then merged: weighted box fusion [7] joins boxes at IoU 0.55, and Gaussian soft-NMS [8] reduces the scores of overlapping fused boxes rather than removing them. Each box retains the scores from the images that found it.

The selected map must govern both the crops and the detections. The detailed pass is therefore repeated on a page otherwise read from R1 alone, or when at least a tenth of the second view's coarse centres move by a quarter letter height between the fitted and selected maps (Appendix A.1 counts these pages).

The detector's raw score is not a probability. We calibrate it against observed matches using a monotone curve (isotonic regression [9]). Its output, p_i, estimates the probability that the box wins a one-to-one match to an annotated letter at rotated IoU at least 0.5. The curve is fitted using held-out detections from the other folds. The reader receives p_i as evidence; no box is discarded for a low p_i. The public records contain 87,494 detections for 62,019 annotated letters of R1: 1.41 proposals per letter. At least one proposal overlaps 99.8% of the letters at IoU at least 0.3. Discarding boxes below $p_i = 0.5$ would reduce that coverage to 95.0%. That loss cannot be repaired by a later language model.

6. Reading a mark together with its neighbours

6.1 The picture and the place

A token's picture is a square, 1.6 times the longer side of its detected box, centred and rotated with it, warped to 224×224 pixels. It is cut from the relit page at available resolution; the relief operations never enlarge the whole scan to manufacture finer detail. The crop includes enough surrounding paper that the reader can see a partial neighbouring stroke or the edge of damage.

An image encoder, DINOv3 ViT-S/16 [3], converts each crop into a numerical description. We train it further on squeeze images, together with the rest of the reader. We also supply 18 measurements describing the box and page: detection scores, size, tilt, overlap with other boxes and alignment quality. The same-looking crop can then be judged differently if it duplicates a stronger box, is unusually large, or lies away from the local text pattern.

6.2 Looking at neighbouring marks

Each token exchanges information with its 32 nearest tokens, counting itself. Distance across a row counts twice as much as distance along it, so this neighbourhood usually includes part of the same row and adjacent rows. The network learns how much weight to give each neighbour from

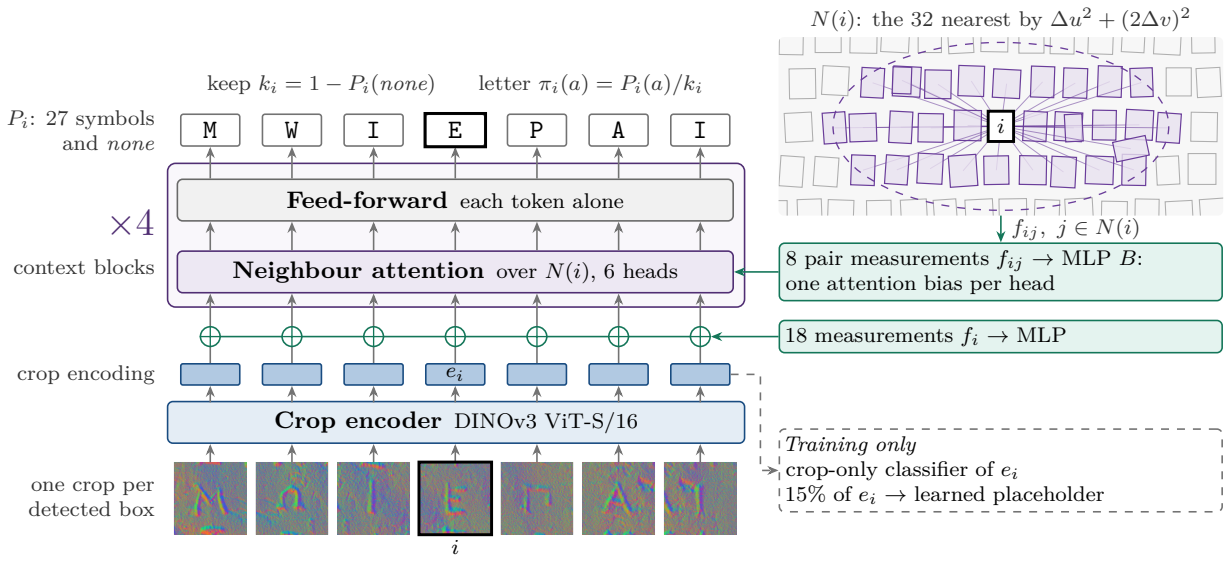


Figure 7: The page network. Seven tokens of public squeeze IG_II2_493_Copy1 (Figure 3, stage 6) pass up through the network; the outlined one is token i. Blue marks the picture, green the place and purple the exchange between neighbours. The crop encoder sees each crop alone; an embedding of the 18 box and page measurements f_i is added to its encoding e_i. In each of four context blocks, every token attends to its neighbourhood $N(i)$, with a bias computed from the eight measurements f_{ij} of each pair, and is then transformed on its own. Top: the reader's first-choice letters for these tokens. Right: $N(i)$ on a schematic page. Distance across rows counts twice, so the dashed ellipse through the 32nd-nearest token is twice as long as it is high. Grey dashed parts act only in training.

both its appearance and its relative position. A box on the left need not matter in the same way as one above, and a duplicate need not count as another letter.

Four successive context blocks repeat this exchange. The network reads the whole page, but each exchange is local; information can travel farther through successive neighbours. Figure 7 shows the whole network; Appendix A gives its equations and measurements.

From the network's probabilities we obtain two quantities:

$$k_i = 1 - P_i(none), \qquad \pi_i(a) = P_i(a)/k_i. \tag{5}$$

$P_i(a)$ is the network's probability for symbol a. Thus k_i asks whether the token should represent a letter; $\pi_i(a)$ asks which letter, conditional on its being one. The reader may favour O over every other letter yet still judge *none* more likely. Conversely, it may be certain that a mark is a letter but unsure which one. The decoder needs both quantities.

6.3 How it learns rejection, duplication and context

Training uses detected boxes, not perfect letter crops. Against the annotation of R1, a one-to-one assignment first matches as many letters as possible at rotated IoU at least 0.3; ties prefer higher calibrated detector probability, then higher IoU. A matched token takes its letter. All others take *none*, including extra boxes on a letter already claimed by another token. Rejection therefore means *this token should not emit an annotated letter*, not necessarily *the paper contains no letter here*.

An extra classifier used only during training learns to read each crop on its own. It receives a letter label for a one-to-one match at IoU at least 0.5, *none* when the best overlap is below 0.3, and no label between those thresholds. This gives the image encoder a direct learning signal.

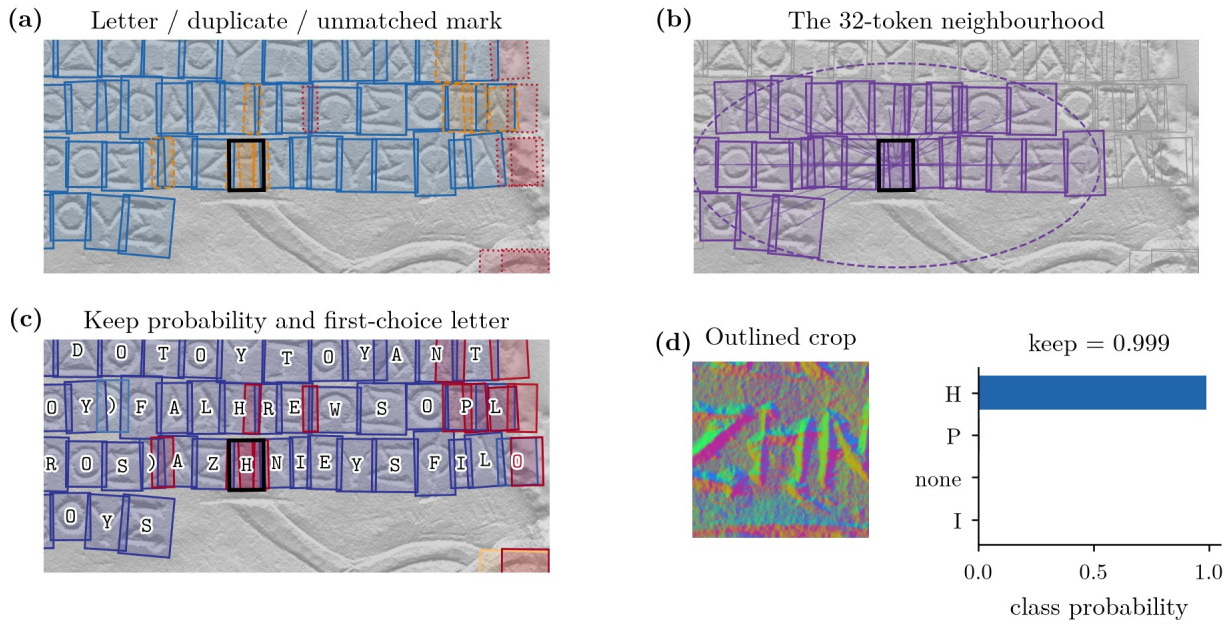


Figure 8: Reading a crop in its neighbourhood. A region of IG_II2_1973a; the black box marks an annotated `H`. (a) Training labels distinguish letters, duplicate boxes and unmatched marks. (b) The 32 tokens available to this box, including itself. Lines join it to each neighbour; learned attention weights are not drawn. (c) Keep probabilities and first-choice letters; red letters disagree with the training label. (d) The outlined crop and its probabilities, including *none*.

The main reader must also learn to use context. We hide the crop description for 15% of the tokens being trained, replacing it with a learned placeholder while leaving their positions and neighbours available. The crop-only classifier still sees the image; the main reader must work without it. Only the main reader's contextual probabilities are used to produce transcripts.

7. Training for changes in light and paper

A network trained on fixed crops could learn one lighting rig or one paper texture too closely. Training therefore varies the relief itself as well as image brightness. We turn the virtual lights, change the apparent depth and fade one view's contribution. Blur and noise alter the texture; small crop shifts and flattened patches imitate imperfect centring and local damage. These transforms vary the image evidence; they do not simulate how a squeeze is made. Appendix A gives their ranges.

The transforms keep a token's class and its page geometry unchanged. The page-wide lighting change is shared within a training window; crop shifts and local damage vary between crops. Figure 9 shows sampled versions of real crops.

Training uses windows of 384 nearby tokens. The nearest 192 contribute to the training error; the others supply context. Two windows make one training step. The two readers train for 5 and 15 *epochs*, or passes through the training data, and both use their final weights. Optimiser settings are listed in Appendix A.

When producing a transcript, the network reads the whole page rather than training windows. Each network sees five crop versions: unchanged, rotated by $-2.5°$ or $+2.5°$, and scaled by 0.95 or 1.05. Geometry and neighbour support stay fixed. The ten distributions are combined by averaging log probabilities and renormalising:

$$\log P_i = \log \operatorname{softmax}\left(\frac{1}{10}\sum_{r=1}^{10} \log P_i^{(r)}\right). \tag{6}$$

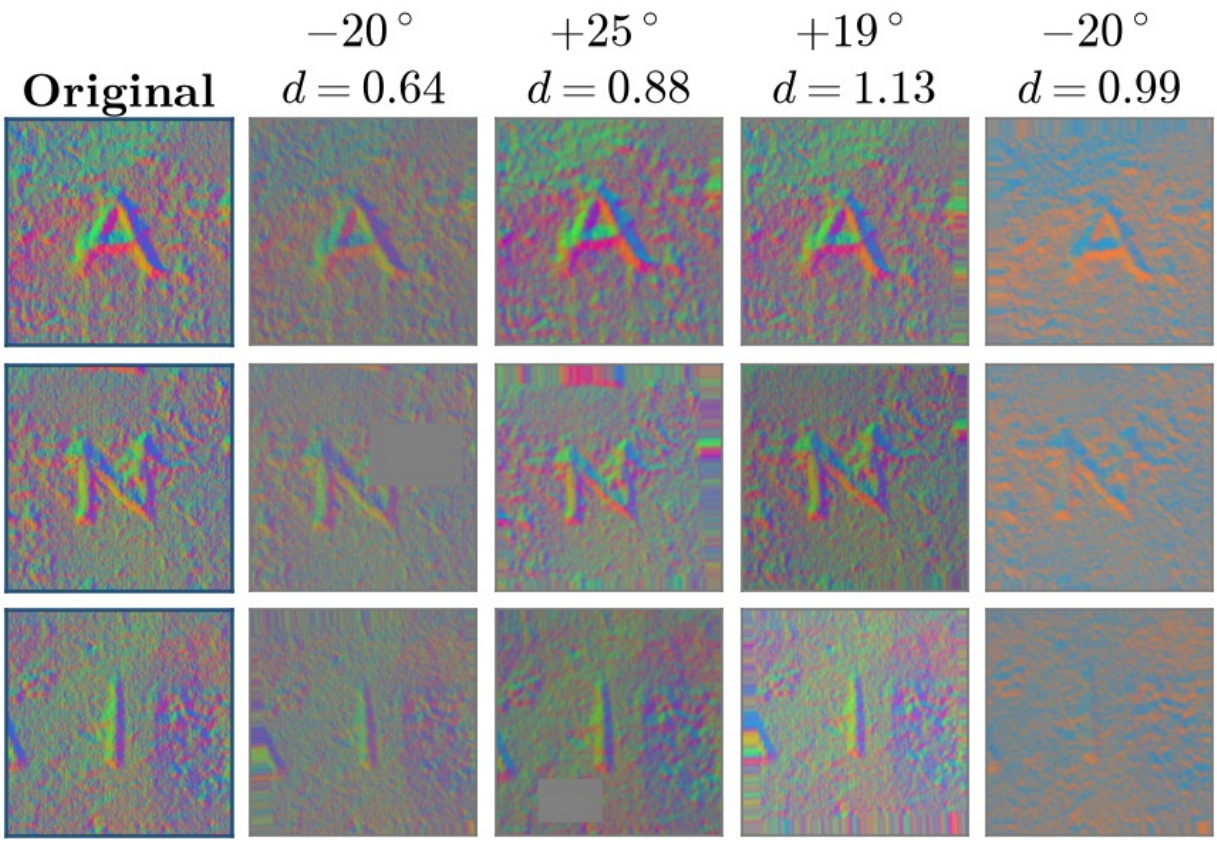


Figure 9: The same letter under different simulated conditions. Three crops of public squeeze IG_II2_493_Copy1 and four sampled training versions of each. Headings give light rotation and depth multiplier; blur, noise, shifts and local damage also vary.

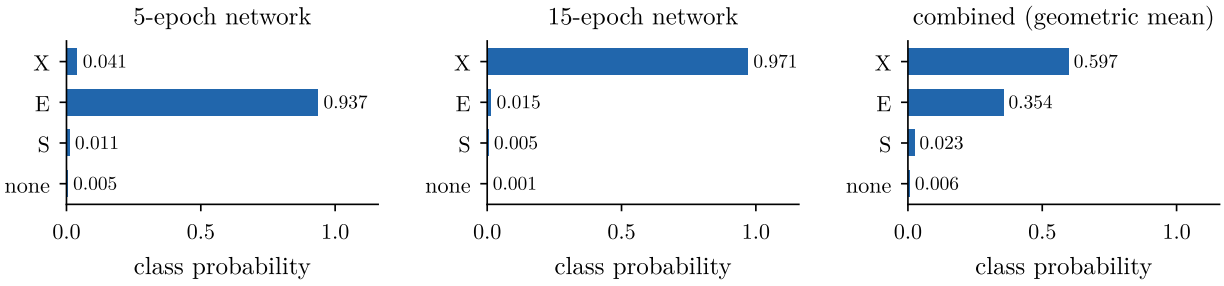


Figure 10: Combining probabilities, not first choices. Token 128 of IG_II2_1973a. The two networks favour E and X; their combination favours X. Each network's distribution already combines five crop views. This is the most strongly disagreeing retained token among those with different first choices. The four classes with greatest summed probability are shown.

This is a normalised geometric mean of the ten distributions: a letter strongly opposed by one version is penalised more than under an arithmetic average.

8. Recognition errors and confidence

Over the public set, the 87,494 tokens include 61,854 matched to an annotated letter and 25,640 assigned *none*. At $k \geq 0.5$, the reader keeps 62,290: it loses 2,269 matched letters (3.7%) and keeps 2,705 unmatched tokens (4.3% of those kept). These counts describe individual boxes. Transcript errors also depend on letter identity, reading order and row boundaries.

The keep probability separates matched letters from the rest better than the calibrated detector score (Figure 11). It is close to observed frequencies at the extremes but underestimates letter

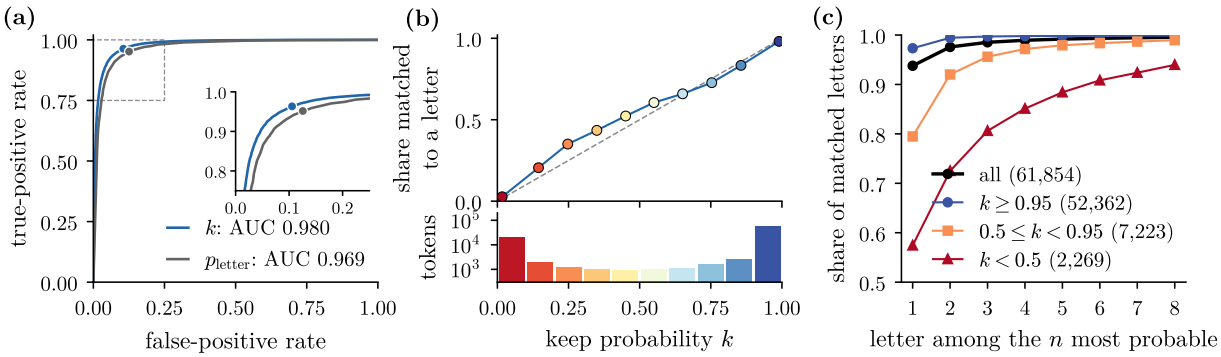


Figure 11: Keeping a box and identifying its letter are different decisions. All public tokens, with each squeeze held out. (a) True- versus false-positive rates as the threshold changes. The area under the curve is 0.980 for the reader and 0.969 for the detector; larger is better. Dots mark threshold 0.5. (b) Observed match frequency against keep probability, with counts below. (c) The fraction of matched letters found among the top n identities, excluding *none*.

frequency in parts of the middle range. Consequently, a probability such as 0.3 should not be treated as a guaranteed frequency, and thresholding at 0.5 loses letters that search could still use.

Among tokens matched to letters, the correct identity ranks first for 93.8%, and among the eight letters offered to search for 99.5%. Many mistakes are small changes in a worn stroke: A becomes Λ, Θ becomes O, or Δ becomes A (Figure 12). The high top-eight rate motivates keeping alternatives. It does not imply that the decoder knows which alternative is correct, nor include letters with no proposal.

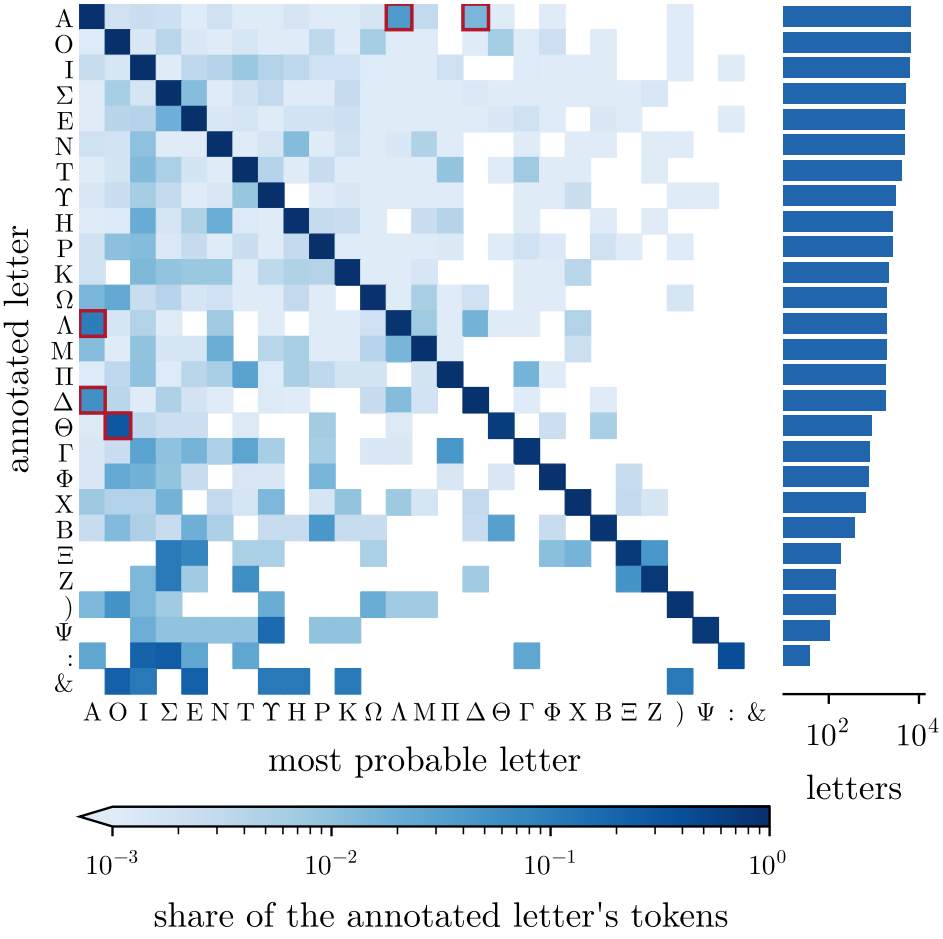


Figure 12: Which letters are confused? The reader's first choice for 61,854 matched public tokens, with letters in Greek. Each row is an annotated letter; each column is a predicted letter. Colour shows the fraction within that row on a logarithmic scale; blank cells have no examples. The side bars show letter counts. Boxes mark the five most frequent mistakes; Θ becomes O in 29.4% of matched Θ tokens. Rejected boxes are measured separately in Figure 11.

9. Assembling rows and choosing their text

The page network has no transcript output. Its boxes must first be put into reading order. Tokens with $k \geq 0.5$ determine the page angle and are grouped by the organisers' version-2 row procedure [2]. A token with $0.05 \leq k < 0.5$ can join the nearest row if it lies within half a letter height of the median baseline and within the row's span expanded by 2.5 letter heights at each end. The others are omitted, including every token below 0.05. Tokens within a row are ordered along the deskewed horizontal direction.

On the running squeeze, 422 strong tokens form 23 rows; 29 weak tokens join them, while 34 join none (Figure 13). The label and much of the ornament are rejected before search. A wrong row grouping can nevertheless merge columns or break a curved row; correct individual letters do not by themselves determine a correct transcript.

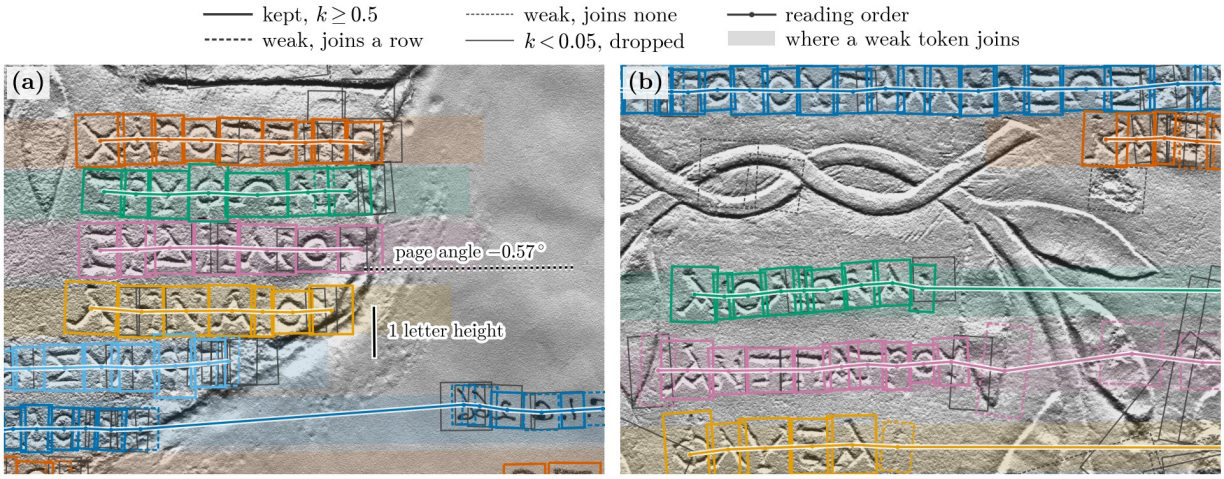


Figure 13: Where a weak mark is allowed to become a letter. Two regions of IG_II2_1973a. Solid coloured boxes have $k \geq 0.5$ and establish rows. Weak boxes joining a row are dashed in that row's colour; weak boxes joining none are dashed grey, and boxes below 0.05 thin grey. Tinted bands show the joining regions, and lines join tokens in reading order. These rows are built from the reader's outputs, without the annotations. Even a weak mark can reach search if it lies on a strong row; an isolated weak mark cannot.

9.1 Comparing possible readings

Each row position offers its eight most probable letters under π_i, and may be skipped if $k_i < 0.95$. For a candidate y emitting positions E from all row positions T, the final score is

$$S(y) = \underbrace{\sum_{i \in E} \log \pi_i(y_i)}_{\text{letter identity}} + w \underbrace{\left[\sum_{i \in E} \log k_i + \sum_{i \in T \setminus E} \log(1 - k_i) \right]}_{\text{emit or skip}} + \lambda L(y) + \beta |y|. \tag{7}$$

The last terms are a character-model score and a per-letter bonus. The bonus counters the fact that adding letters normally adds negative log probabilities. At $k_i \geq 0.95$, emission is compulsory and the keep term is set to zero. Elsewhere k_i is clipped to $[10^{-4}, 1 - 10^{-4}]$ for stable logarithms. The final weights are $(\lambda, w, \beta) = (0.5, 1, 1)$.

The search proceeds left to right, keeping the 256 highest-scoring partial readings at each step (*beam search*). It runs at five settings of (λ, w, β): $(0, 1, 0)$, $(1, 1, 0)$, $(1, 2, 2.5)$, $(1, 2, 3.5)$ and $(1, 3, 3)$. Their union gives a finite candidate pool, covering different compromises between letter identity, retention and text plausibility. A later scorer can reorder this pool, but cannot expand it. Search cannot recover a letter if no relevant token offers it among its top eight alternatives.

9.2 Breaking context at gaps

The median spacing of neighbouring strong tokens estimates the usual distance between letters. If two consecutive emitted tokens stand more than 1.6 times that distance apart, the character models start afresh. Thus a name before a broken stretch is not treated as directly adjoining a word after it. This reset adds neither guessed letters nor a special symbol to the final transcript. Row boundaries also restart context.

10. Using Greek letter patterns

10.1 Patterns learned from existing texts

Some letter sequences are more plausible in Greek than others. The character models estimate this plausibility one letter at a time; they do not consult a dictionary of complete inscriptions.

The public row transcripts contain 62,246 letters. We count their short character sequences using *prediction by partial matching*, PPM-C [10, 11]. The stored model supports up to 11 preceding letters; decoding uses at most four. A second model, with up to five preceding letters, is built from 178.1 million letters of openly licensed external Greek: Perseus, First1KGreek, DDbDP, I.Sicily, IAph2007 and IOSPE (CC BY or CC BY-SA; manifests record URLs, licences and pinned revisions). Before any counting or training, this text is decontaminated (Section 10.2).

PPM-C predicts from the counts following a given sequence. When those counts do not cover a letter, it looks at a shorter sequence, excluding letters already accounted for; with no evidence left, it uses a uniform distribution. The public and external predictions are mixed:

$$P_{\text{base}}(a \mid c) = (1 - \omega)P_{\text{public}}(a \mid c) + \omega P_{\text{ext}}(a \mid c), \tag{8}$$

with $\omega = 0.6$ (Section 12). The public model excludes the fold being evaluated. For test reading, it uses all 224 public squeezes.

10.2 Removing the evaluated inscriptions from the external text

External text can hold the inscriptions being evaluated: an edition of a squeeze among other inscriptions, or a quotation of it in a literary text. The external text is therefore filtered against the transcripts of all 274 annotated squeezes, the 224 public and the 50 TROGS-26 test squeezes, using every annotated letter of both views. This is the one use of the test annotations: they decide which external text is removed (Section 12).

A 15-letter sequence of a squeeze is *rare* if at most 10 external documents hold it. A rare sequence is text of one inscription: its editions, copies and quotations. Formulae, titles and common names recur in more documents and stay. An epigraphic document is taken to be a squeeze's inscription if it holds at least 5 distinct rare sequences of the squeeze, making up at least 4% of the sequences of the shorter of the two texts. These documents are removed whole: 7, found for 5 public and 0 test squeezes. In every other document, each occurrence of a rare sequence is cut out and its line split there, so neither the counts nor the neural model's training windows hold it: 38,827 letters from 1,723 documents. A list naming the same person, or a decree repeating a clause, thus loses only the shared stretch. A final scan finds no rare sequence of any squeeze in any remaining line.

Both thresholds were set on public squeezes only, by one measurement on the PHI Greek inscriptions [13], which hold editions of most public squeezes under the IG numbers of their names. PHI's terms do not allow redistribution; it is read for this measurement alone, and no model holds its text. Counting documents over the open corpora and PHI's 173,677 records together, 91.5% of the sequences of a public squeeze found in a record citing its IG number (296 records of 149 squeezes) occur in at most 10 documents; the rest are formulae. Of the 229 records holding at least 5 rare sequences of their squeeze, none shares less than 5.0% of the shorter text. Without PHI, no sequence is held by more documents, so every sequence rare in this measurement is rare in the filter. Text that an evaluated inscription shares with more than 10 other documents stays: it is as available to a reader of any new inscription.

10.3 Repetition on the same squeeze

Names and formulae can repeat within one inscription even when uncommon elsewhere. A first pass selects draft rows at $(\lambda, w, \beta) = (0.5, 2, 2)$. For each row, a temporary PPM-C model, called the *page cache*, counts sequences in the other draft rows, using up to four preceding letters and

restarting at gaps. The second pass uses

$$P_{\text{cachemix}} = 0.8P_{\text{base}} + 0.2P_{\text{other rows}}. \tag{9}$$

The row being read never supplies its own cache. This prevents direct self-counting, but errors can still pass between rows: a mistaken draft repeated elsewhere can favour the same mistake. When there are no other non-empty rows, no cache mixture is made.

10.4 Using the letters on both sides

A complete candidate can be scored using the letters after a doubtful character as well as those before it. Forward and backward PPM models use the same corpora in opposite directions; cache mixtures are made on both sides. For each gap-delimited run,

$$L(y) = \sum_i \left[\log P_{\text{f}}(y_i \mid y_{<i}) + \log P_{\text{b}}(y_i \mid y_{>i}) - \log P_{\text{f}}(y_i \mid \varnothing)\right]. \tag{10}$$

Subtracting the context-free term avoids counting a symbol's base preference twice. The sum serves only to rank candidates and is not normalised over strings. The beam still proposes left to right; right context enters when completed candidates are scored.

10.5 Adding longer context

A neural character model can use a longer stretch of preceding text: up to 192 symbols. It first learns from the decontaminated external text, then receives further training on public transcript rows mixed with external sequences. Training for a held-out fold excludes its transcripts; the test model uses all public rows. Many training sequences start within a line, so the start symbol means that the preceding text is unknown; a word need not begin there.

Only the forward term in the *final* score changes:

$$\log P_{\text{f}}(a \mid c) \quad \longmapsto \quad \log[(1-\nu)P_{\text{f}}(a \mid c) + \nu P_{\text{neural}}(a \mid c)], \tag{11}$$

with $\nu = 0.8$ (Section 12). The backward and subtracted context-free terms stay unchanged. The neural model supplies neither the beams nor the draft rows for the cache. It reads each candidate run from a start symbol after every gap; it can only choose a text already present in the finite visual candidate pool.

10.6 The order of the passes

Decoding a squeeze takes two search passes over its rows and one final choice:

1. *Draft pass.* Beam search at the five settings of Section 9, with P_{base}, gives a candidate pool. Its best candidate under Eq. (7) at $(\lambda, w, \beta) = (0.5, 2, 2)$, with the two-sided L of Eq. (10) over P_{base}, is the row's draft.
2. *Cache pass.* The other rows' drafts fill the page cache; beam search at the same five settings, now with P_{cachemix}, gives the row's final pool. A row whose other rows are all empty keeps its draft pool.
3. *Final choice.* Each candidate in the final pool is scored by Eq. (7) at $(\lambda, w, \beta) = (0.5, 1, 1)$, with the two-sided L over the cache mixtures and its forward term mixed with the neural model (Eq. (11)). The best candidate is the row's text.

11. Combining five readings

The five fold detectors are different models. On a new squeeze they can propose different boxes, estimate different scales and fit different registrations. At test time the *same* 5- and 15-epoch page

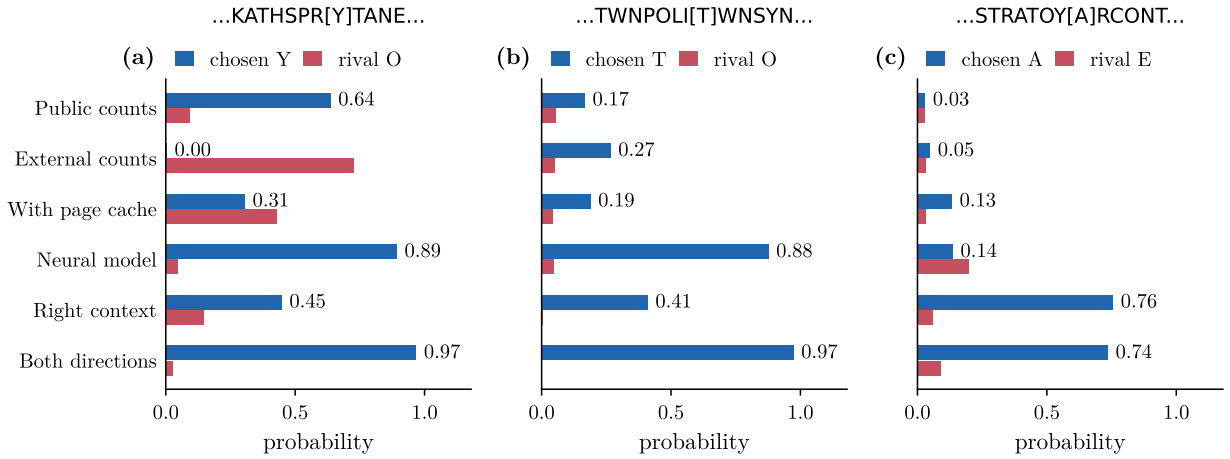


Figure 14: What text context favours. Three positions in public IG_II2_493_Copy1, comparing the chosen letter with one rival. Each stage adds one part of the adopted text model: external counts, the page cache, the neural model and right-hand context, on top of the public counts; the six rows are the same in all three panels. The two-sided scores are normalised across letters for display. These plots measure text prediction alone; transcript accuracy is measured in Section 12.

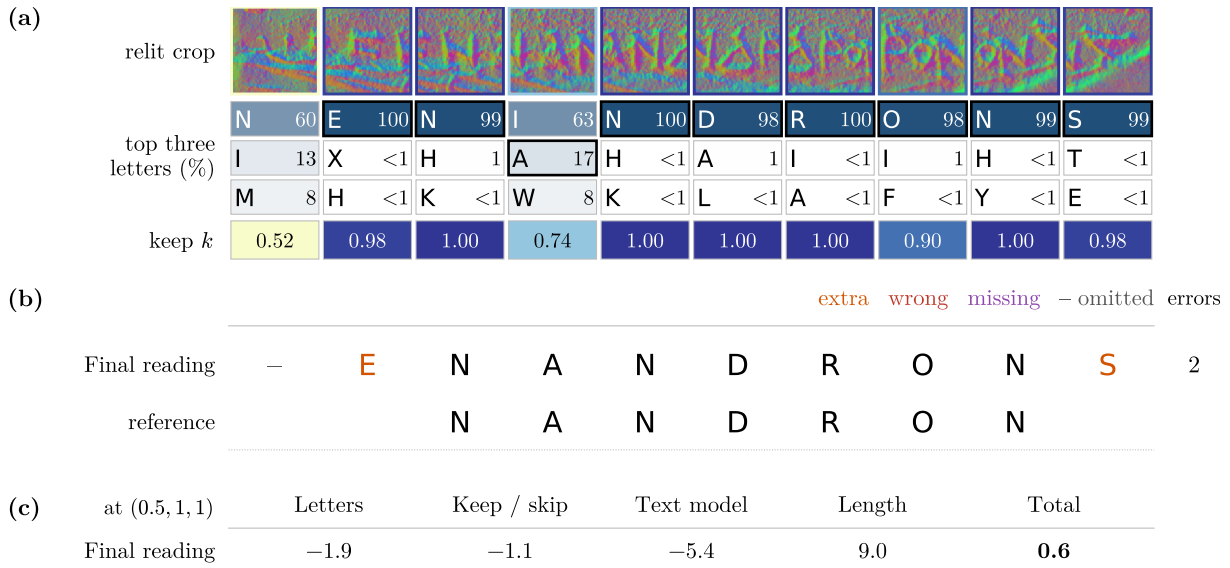


Figure 15: Choosing a better reading, with errors still visible. Row 23 of IG_II2_1973a. (a) Ten crops, their three most probable letters and keep probabilities. Boxes mark the selected letters. (b) The final reading and the reference. The initial N is skipped and I changes to A; the extra E and S remain. (c) The parts of the final reading's score at the adopted weights $(\lambda, w, \beta) = (0.5, 1, 1)$: letters, keep/skip, text model and length. This example is a row of 10–25 positions.

networks, trained on all 224 public squeezes, read every chain's page. The five texts therefore differ mainly through detection and alignment.

We combine the transcripts with ROVER [12]. First, choose the transcript requiring the fewest edits in total to match the other four. This is the *pivot*. Align the other transcripts to it, treating line breaks like other symbols. Then count the votes at each aligned position:

- The most frequent symbol wins. A gap counts as one vote, just like a letter.
- A tie keeps the pivot's symbol.
- A string inserted between pivot positions needs at least three identical votes and must beat the votes for no insertion.

Winning gaps produce no character, and empty output rows are removed. The two highlighted positions in Figure 3 show a tie resolved by the pivot and a position removed by a gap vote.

Across the test squeezes, two chains' transcripts differ by an average of 5.0% of the longer transcript's length, measured by edit distance. The vote changes 38 output characters relative to the pivot (13 substitutions and 25 inserted characters) and removes 31 pivot characters. These changes measure disagreement between chains; a misreading shared by most chains survives the vote.

12. How well it reads

12.1 Public development and the fixed test output

There are 224 public squeezes: 181 training, 22 validation and 21 internal-test. The five detectors train on the 181 training squeezes only, each without its own fold; validation and internal-test pages are held out from every detector. Reader folds have sizes 45, 45, 45, 45 and 44, each containing its corresponding detector fold plus eight or nine additional public squeezes. A public squeeze is therefore read by a detector that did not train on it, and by the reader pair, public language model and neural fine-tune that exclude its entire reader fold. The pair reads through the map selected on the images (Section 3), although its training crops were cut through the map fitted to detected letter centres. The pair that reads the test pages trains in the same way on all 224 public squeezes.

For each reported fold, the weights of the text models and of the decoder are selected using the other four folds over 1800 points: the external weight $\omega \in \{0.3, 0.45, 0.6\}$, $\lambda \in \{0.3, 0.4, 0.5, 0.65, 0.8, 1.03\}$, $w \in \{1, 2, 3, 4\}$, $\beta \in \{0, 1, 2, 3, 4\}$ and the neural weight $\nu \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$. This keeps each fold out of its own weight selection; the public rates of Table 1 read each fold at its own choice. The point with the fewest errors over all five folds, $\omega = 0.6$ and $(\lambda, w, \beta, \nu) = (0.5, 1, 1, 0.8)$, is the one used on the test pages (Figure 16); it lies at the largest external weight of the grid. The choice made without a fold matches it in three of the five folds; over all folds it makes 12,565/131,473 errors (9.56%), a figure selected on those same squeezes. Other choices are shared by all folds. The architecture and training variation were developed using public results, and the detector epoch was chosen once, on fold 0; the public estimate is not an independent test of those choices.

The organisers' version-2 scorer [2] compares one output transcript with each annotated view and reports three rates. Core CER counts edit operations against confident letters, including row-boundary characters; rows with no confident letters are omitted. All CER counts them against every annotated letter. Opt CER lets uncertain letters be omitted at no cost; an uncertain letter read correctly joins the reference count. Missing, substituted and inserted letters, and row splits or merges, all cost. We sum errors and reference characters over both views, rather than averaging per-squeeze percentages. All counts in Table 1 follow that definition.

Evaluation set	Errors	Characters	Core CER
Public (224 squeezes)	12,612	131,473	**9.59%**
Without `Merged` (208)	7,935	96,628	8.21%
No related squeeze in another fold (176)	10,631	105,068	10.12%
TROGS-26 test (50 squeezes)	1,464	15,862	**9.23%**

Table 1: Character error rates against both annotated views. Public squeezes use one held-out chain each, and each fold is read at the weights chosen on the other four; test squeezes use readers trained on all public data and a vote over five chains. Public fold rates (0–4): 8.73%, 9.92%, 12.87%, 9.01%, 6.95%.

The ROVER transcripts of the 50 TROGS-26 test squeezes [2] were frozen before the reported final evaluation. It gives 1,464/15,862 Core errors (9.23%), All CER 2,128/16,944 (12.56%), and

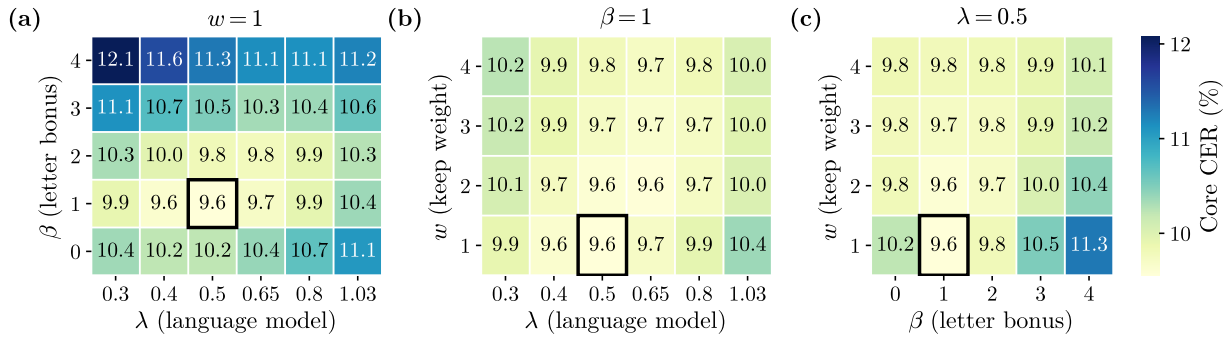


Figure 16: What the decoding weights change. Three slices of the public grid through the selected point, at its external and neural weights: (a) language-model weight and letter bonus; (b) language-model and keep weights; (c) letter bonus and keep weight. Cells show Core CER (%), rounded to one decimal place. Each setting chooses among saved candidate readings; it does not run the visual models again. Each held-out fold chooses its weights using the other folds; three of the five select the boxed point. The plotted cells themselves are pooled over all public squeezes, for diagnosis.

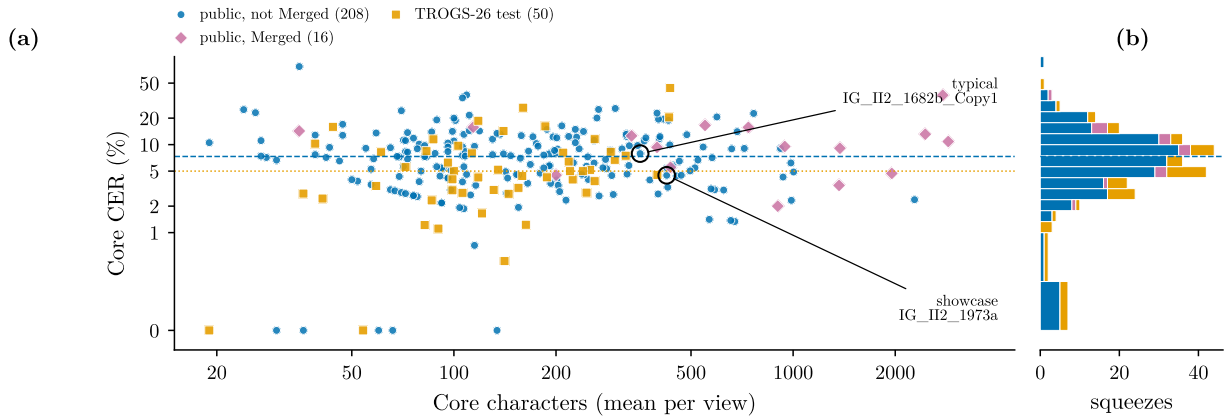


Figure 17: Some inscriptions are much harder than others. (a) Each public and test squeeze's Core CER against its mean reference length per view. Public points are split by the filename marker `Merged`; test points use the frozen five-chain score. Lines mark public and test medians. (b) Counts of squeezes in error-rate bands. CER axes are linear below 1% and logarithmic above it, so true zero-error cases are retained rather than shifted to an artificial floor.

Opt CER 1,282/16,052 (7.99%). All counts include both annotation views, with a prediction for every squeeze. Transcript hashes were recorded with the score, and no model, setting or transcript changed in response to that evaluation. Test labels entered one step, the decontamination of the external text (Section 10.2): they decided which external text was removed before training. Otherwise no test label entered training, the folds or parameter selection.

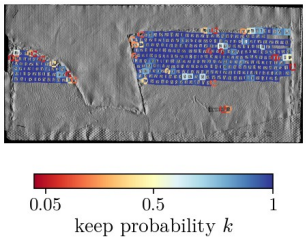
12.2 The average hides difficult squeezes

The public median per-squeeze Core CER is 7.34%, below the pooled 9.56%. The 22 worst public squeezes account for 34.1% of the errors in 12.1% of the reference characters. The 208 squeezes without `Merged` in their names make 7,935/96,628 errors (8.21%); this subset is reported alongside the full benchmark. The 50 test squeezes range from 0.00% to 44.0%, with median 5.00% (Figure 17).

12.3 A complete example

The running squeeze is useful for showing ornament and crops, but is easier than average. For a fuller error example, we select the public squeeze with 150–450 Core characters in R1 whose pooled CER is nearest the median within that size range: IG_II2_1682b_Copy1 (7.91%). Figure 18

shows every row. Its R1 comparison has 28 errors in 354 characters. Lost letters occur at row starts and ends as well as inside words, and some rows begin with a letter the reference lacks. The annotations, proposal set, reader and row search all constrain what can be read; the language model has not completed the words to familiar forms.



Row	Reference	System
1	TONAI	EEPITONIA
2	ERAMWSAIKATATAGEGRAMMENA	ERAMWSAIKATATAGEGRAMMENA
3	ISTYLIA:DDP;ARIQMOSEPITONAIWNWNDE	STYLIA:DDPARIQMOSEPITONAIWNWNDE
4	AIEGDYOINTETRAGWNOINEKASTON;	EGDYOIHTETRAGWNONEKASTONP
5	TASKAIKALYMMATADDITAYTAOMISQWSAME	TASKAIKALYMMATWDDTAYTAOMISQWSAME
6	OSKAIKOLLANWMOBOIONKAITALLAWNANDEHIPRO	NOSKAIKOLLANWMOBOIONKAITALLAWNANDEIP
7	THNERGEUINKAITHNALIFHNTWNXYLWNKAITHN	THNERREUINKAITHNALIFHNTWNXYLWNKAITHN
8	DORWSINLHNKERAMOYAPODWSEIEXERGASAMENOS	DORWSILHNKIRAMOYAPODWSEIEXERGASAMENOS
9	PIDIOTIMOYYNICIWNIKAIQARGHLIWNIKAISKIROFORIW	SPIDIOTIMOYYNICIWIKAIQARGHLIWNIKAISKIROFORIW
10	MISQWTHSAGAYSALAIEYS:CC:EGGYHTAI:CARIAS	AISQWTHSAGAYSALAIEYS:CC:EGGYHTAISARIAS
11	PEIQIOYFREARIKLEOYSALAIEYS	PEIOIOYFREARIKLEOYSALAIEYS

Figure 18: A complete transcript and its errors. Held-out public squeeze IG_II2_1682b_Copy1, selected by the size-and-median rule in the text. Left: boxes available to the decoder (solid strong, dashed weak). Right: the R1 Core reference and system output. Red marks substitutions, purple missing letters, and orange extra letters; all are underlined. An output row without a reference number is extra. The 28 errors include row-boundary edits.

13. Limits of the reading

A missed proposal is irrecoverable. A letter below keep 0.05, or a weak token outside every strong row, is lost before character-model scoring. A mistaken row order can mix columns even when every crop is legible. Each context block of the page network sees only a local neighbourhood; repetition elsewhere on the page reaches the decoder through the draft rows, which can repeat errors as readily as names.

Relief recovery fails where shadows do not behave like directional slopes, scans cover different paper, or registration fails at a seam. Local agreement weighting limits contamination but cannot recover a stroke hidden from the remaining light. The three virtual channels re-present two measured slopes: they add no observation and recover no absolute surface depth.

A high keep probability can also protect a wrong mark. At or above 0.95, search cannot skip it; below that, a large skip cost may still outweigh a text-model penalty. Conversely, a visible letter omitted from a confident-letter reference can count as an insertion. The training labels teach the network to match annotation decisions as well as letter shapes. Neither visual plausibility nor agreement among five related chains proves a reading correct. The transcript is a proposed reading to check against the photographs, not a restoration or a scholarly edition of the inscription.

A. Implementation details

A.1 Alignment and detection

Across the 224 public squeezes, the number with at least 98% of annotated letter correspondences within a quarter letter height rises from 168 for the centre fit to 197 for the selected map (Figure 4c). The rule of Section 5 repeats detection on 18 public squeezes and on 52 of the 250 test-chain records (five chains for each of 50 test squeezes). Table 3 lists the registration search.

A.2 The page network

The image encoder has 12 layers of width 384, six attention heads, a class token, four register tokens and 14×14 image patches. The class token c_i and mean patch token $\bar{z}_i$ (excluding registers) form a 384-dimensional crop encoding e_i. The 18 measured features f_i are standardised using training-fold statistics, then embedded and added:

$$e_i = \mathrm{LN}\big(W_e[c_i; \bar{z}_i]\big), \qquad x_i^{(0)} = e_i + W_2\,\mathrm{GELU}(W_1 f_i). \tag{12}$$

LN denotes layer normalisation and GELU the activation function. The neighbourhood $N(i)$ contains the 32 nearest tokens, including i, under $\Delta u^2 + (2\Delta v)^2$, where u runs along rows and v across them in letter heights. For attention head η, the weight on neighbour j is

$$\omega_{ij}^{\eta} = \underset{j \in N(i)}{\mathrm{softmax}} \left(\frac{q_i^{\eta} \cdot k_j^{\eta}}{\sqrt{64}} + B(f_{ij})_{\eta} \right), \qquad B : \mathbb{R}^8 \to \mathbb{R}^{64} \to \mathbb{R}^6. \tag{13}$$

Here q_i^{η} and k_j^{η} are learned query and key vectors, distinct from the keep probability k_i; f_{ij} contains the eight pair measurements in Table 2. Four blocks combine the six heads, with residual connections and a $384 \to 1536 \to 384$ feed-forward network. The output is $P_i = \mathrm{softmax}(W_h\,\mathrm{LN}(x_i^{(4)}))$. The reader has about 29.2 million parameters: 21.6 million in the image encoder and 7.1 million in the context blocks.

Input	Measurements
Detection (5)	Logit of calibrated probability; fused score; scores from the three renderings
Box shape (5)	Log height, width and aspect ratio; sine and cosine of twice the box's relative tilt
Overlap (3)	Maximum IoU; maximum IoU with a more probable neighbour; log overlap count
Page (5)	Log letter height in pixels; log token count; confident-token fraction; single-view flag; registration inlier share
Token pair (8)	Along-row and across-row displacement; log distance; IoU; log height and width ratios; probability-logit difference; self indicator

Table 2: Measurements supplied with each crop. Positions and sizes use letter heights in a page frame rotated to the estimated text direction. Overlap features use the 16 nearest boxes, approximated as rectangles in that frame. The page angle is estimated from detections before reading, then re-estimated from keep probabilities when forming rows.

A.3 Training objective and settings

With core tokens C and auxiliary-labelled core tokens R, the training loss is

$$\mathcal{L} = \frac{1}{|C|} \sum_{i \in C} -\log P_i(t_i) + \frac{0.5}{|R|} \sum_{i \in R} -\log P_i^{\mathrm{read}}(t_i^{\mathrm{read}}). \tag{14}$$

Here t_i is token i's training label, a letter or *none*, and t_i^{read} its label for the crop-only classifier, whose probabilities are P_i^{read} (Section 6). Context tokens outside the core also propagate gradients

through their crops. Two windows are accumulated per optimiser step. AdamW uses learning rates $5 \cdot 10^{-5}$ for the encoder and $5 \cdot 10^{-4}$ for the rest, weight decay 0.05, 5% linear warm-up and cosine decay. The 15-epoch reader uses encoder stochastic depth 0.1; neither reader selects a checkpoint on a reader-validation split. The separate neural character model has about 14.25 million parameters.

The relief transforms turn the virtual lights by up to 30°, scale each gradient component within $\exp(\pm 0.2)$, and fade one component in 30% of training windows. Apparent depth ranges from 0.6 to 1.3 times its original value, blur reaches one pixel and noise 0.02. Each crop can move by 4% of its side; with probability 0.15, a rectangle covering 3–12% is flattened.

Component	Fixed settings
Registration	Phase correlation over quarter turns and small angular offsets; ECC Euclidean, then affine. Local levels: tiles of 12, 6 and 6 letter heights with displacement limits of 5, 1.5 and 0.5 letter heights; thin-plate spline smoothed until orientation-preserving. Median tile correlation selects one of three maps.
Detector	YOLO26-L OBB; 100-epoch schedule, epoch 59 used. Chain 0 chooses the best held-out $\text{mAP}_{50:95}$ epoch; the same epoch is used for all chains. Detailed inference: 40-pixel letter height, 1280-pixel tiles, 25% overlap; one-to-many detection head with rotated NMS at IoU 0.8 within each tile. Across the three images, weighted box fusion at IoU 0.55, then Gaussian soft-NMS.
Crop encoder	DINOv3 ViT-S/16, 12 layers, width 384, six heads; class token plus mean of 196 patch tokens, projected $768 \to 384$ with layer normalisation.
Page blocks	Four pre-LN blocks, six heads of width 64, 32-token support; pair-bias MLP $8 \to 64 \to 6$ per block; feed-forward $384 \to 1536 \to 384$; dropout 0.1.
Reader training	384-token windows, 192-token core, two windows per step; mask probability 0.15; auxiliary loss weight 0.5; AdamW, bf16, gradient-norm clip 1. Final checkpoints at 5 and 15 epochs.
Crop augmentation	Rotation ±5°, scale 0.9–1.1; brightness gain parameter 0.2 and bias parameter 0.08, each multiplied by a draw in $[-0.5, 0.5]$, plus the relief transforms above. No flips.
Neural text base	Eight layers, width 384, six heads, context 192, tied embeddings; 6,000 steps, 512 sequences per step, AdamW peak $6 \cdot 10^{-4}$, 500 warm-up steps, cosine decay; every 200th external line held out. 80% of sampled sequences start inside a line.
Neural fine-tune	100 steps; per step 64 transcript rows and 64 external sequences, their losses averaged; AdamW at 10^{-4}. Fold-specific transcript exclusions; all-public model for test.
Final selection	Width-256 beams, eight letters per token; gap restart at 1.6 pitches; cache 0.8/0.2; external weight $\omega = 0.6$; $(\lambda, w, \beta, \nu) = (0.5, 1, 1, 0.8)$.

Table 3: Settings that determine the computation. Training labels and the two distinct matching thresholds are described in Section 6; evaluation and selection boundaries are in Section 12.

B. Sources and reproduction

The adopted inference configuration is `configs/pagenet_only_style_neural.json`. It resolves the image-registration/detect-again path, the two all-public mild-style readers, two-sided character models, and the neural forward mixture; `configs/pagenet_only_style_neural_oof.json` applies it to the public folds, each read by the pair trained on the other four. Reader training recipes are `configs/style_mild_5_epochs.json` and `configs/style_mild_15_epochs.json`; public folds are `data/folds_public224.json`. The code and configurations are at https://github.com/ppdms/greek_squeezes_e2e. The trained models are public at https://huggingface.co/

buckets/papadimas/greek_squeezes in the trogs26/ namespace, with a checksum manifest (MANIFEST.sha256).

The code follows the stages described in the report:

- Alignment and detection: trogs26/registration_images.py and trogs26/detect/redetect.py.
- Relief and crops: trogs26/relief.py and trogs26/dino/crops.py.
- Reading, text models and search: trogs26/pagenet/, trogs26/lm/ and trogs26/decode/.
- Transcript voting: trogs26/rover.py.

The reader jobs are gcp/jobs/pagenet_fold.sh (a fold's networks), gcp/jobs/pagenet_all.sh (the networks trained on all 224) and gcp/jobs/pagenet_images_read.sh (reads through the registration chosen on the images, of the public folds or of new pages). The character-model training job is gcp/jobs/neural_lm.sh. For a new split, python -m trogs26.decode.newpages runs variant P with the adopted configuration; python -m trogs26.rover combines its five chain transcripts. paper/data/trogs26_score.json preserves the resolved configuration, the 50 prediction hashes, per-squeeze counts and final totals from the fixed-output score. The paper generators reuse those counts; they read no test annotation and run no scorer.

Figure provenance. paper/make_figures.py writes the figures and their numerical labels to paper/figs/, except the diagram of the page network (Figure 7). It is drawn by hand in TikZ (paper/diagram/pagenet.tex) around the crops and first-choice letters of Figure 3, stage 6; its neighbourhood is a schematic page. Detector, token, row and error plots use the current public out-of-fold records, with the 18 detect-again records substituted where required. The main specimen is IG_II2_1973a; the complete error example is selected by the stated size-and-median rule. Saved public assets from IG_II2_493_Copy1 supply the augmentation draws and page pictures; paper/decode_assets.py recomputes their decode and letter-model distributions under the final system. The neighbourhood picture shows geometric support only. ROVER examples use test *outputs* only. paper/make_results.py writes the result macros and table from stored sweeps and the frozen score. LuaLaTeX builds this manuscript with latexmk -pdf report.tex. Neither figure generation nor manuscript compilation runs a page network or the test scorer.

Acknowledgments

The image encoder uses Meta AI's DINOv3 pretrained weights under the DINOv3 License. The detector uses Ultralytics YOLO26. We thank the TROGS organisers for the photographs, annotations and evaluation tools, and the open Greek text projects for their corpora and explicit licences.

References

[1] Howe, N.R., Chang, F., Falbo, I., Brown, T., Hershkowitz, A.: Character recognition for Greek squeezes. International Journal on Document Analysis and Recognition 28(3), 345–356 (2025). https://doi.org/10.1007/s10032-025-00540-y

[2] Hershkowitz, A., Howe, N., Cosgrove, B., Brown, T.: TROGS-26 test images, with annotations and version-2 evaluation code. Data, Smith College, Northampton, MA (2026). https://scholarworks.smith.edu/dds_data/19

[3] Siméoni, O., et al.: DINOv3. arXiv:2508.10104 (2025).

[4] Jocher, G., Qiu, J., Liu, M., Lyu, S., Akyon, F.C., Kalfaoglu, M.E.: Ultralytics YOLO26: unified real-time end-to-end vision models. arXiv:2606.03748 (2026). Software: https://github.com/ultralytics/ultralytics.

[5] Bookstein, F.L.: Principal warps: thin-plate splines and the decomposition of deformations. IEEE TPAMI 11(6), 567–585 (1989). https://doi.org/10.1109/34.24792

[6] Evangelidis, G.D., Psarakis, E.Z.: Parametric image alignment using enhanced correlation coefficient maximization. IEEE TPAMI 30(10), 1858–1865 (2008). https://doi.org/10.1109/TPAMI.2008.113

[7] Solovyev, R., Wang, W., Gabruseva, T.: Weighted boxes fusion: ensembling boxes from different object detection models. Image and Vision Computing 107, 104117 (2021). https://doi.org/10.1016/j.imavis.2021.104117

[8] Bodla, N., Singh, B., Chellappa, R., Davis, L.S.: Soft-NMS—improving object detection with one line of code. In: ICCV, pp. 5562–5570 (2017). https://doi.org/10.1109/ICCV.2017.593

[9] Zadrozny, B., Elkan, C.: Transforming classifier scores into accurate multiclass probability estimates. In: KDD, pp. 694–699 (2002). https://doi.org/10.1145/775047.775151

[10] Cleary, J.G., Witten, I.H.: Data compression using adaptive coding and partial string matching. IEEE Transactions on Communications 32(4), 396–402 (1984). https://doi.org/10.1109/TCOM.1984.1096090

[11] Moffat, A.: Implementing the PPM data compression scheme. IEEE Transactions on Communications 38(11), 1917–1921 (1990). https://doi.org/10.1109/26.61469

[12] Fiscus, J.G.: A post-processing system to yield reduced word error rates: Recognizer output voting error reduction (ROVER). In: IEEE ASRU, pp. 347–354 (1997). https://doi.org/10.1109/ASRU.1997.659110

[13] The Packard Humanities Institute: Searchable Greek Inscriptions. https://inscriptions.packhum.org